

# What the Lesson Doesn't Carry Home

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*A BrainLift on why piano students don't practice at home.*

## Owner

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## Purpose

The purpose of this BrainLift is to research the science behind why piano students don't practice at home. It starts from three beliefs: students don't know what to practice, they don't know how to practice, and the problem isn't motivation. It's information and skill.

## North Star

If a student knows exactly what to practice and exactly how to do it, does home practice actually happen?

## In Scope

- **What to practice** — why vague assignments fail and what specific instructions actually change
- **How to practice** — the strategy gap: novices default to playing the whole piece start to finish regardless of how much time or motivation they have
- **Why what and how don't reach the bench** — the translation breakdown between lesson and home: parents as unreliable relay, inflated self-reports, no feedback at the moment of practice
- **The motivation counter-argument** — the case that motivation is the real problem, and why the information-and-skill view answers it
- **Consistency over volume** — why short daily sessions outperform long weekly ones at the same total time, why the spacing effect from cognitive science applies to piano, and why the field hasn't tested this cleanly in beginners

- **The cost of getting this wrong** — how unproductive practice becomes the failure spiral that ends in quitting

## Out of Scope

- **Talent vs practice debate** — whether ability is innate or earned. Contested, but doesn't change what students should be doing at home.
- **Instructor quality and pedagogical style** — what makes a great piano teacher. The BrainLift assumes the lesson itself is fine. The breakdown happens after the student leaves.
- **Cost, scheduling, and access barriers** — the sociology of who gets to study piano. Important, but downstream of the practice failure this BrainLift studies.
- **Curriculum and repertoire selection** — which pieces and method books to use. Different conversation. The strategy deficit holds regardless of repertoire.
- **Performance anxiety and stage fright** — real for advanced students, but this population can't get productive practice off the ground in the first place.
- **Professional and conservatoire-level practice** — the high end of practice research. Informative, but the population here is students still learning how to practice, not optimizing it.

## Knowledge Tree

### Category 1: The Information Gap

*The first half of the BrainLift — that students don't know what to practice when they sit down at home, and what changes when the assignment is specific.*

#### **1.1 Supporting — Miksza on specific written practice goals**

**Source:** Miksza, P. (2015). The effect of self-regulation instruction on the performance achievement, musical self-efficacy, and practicing of advanced wind players. *Psychology of Music*, 43(2), 219–243.

**DOK 1 — Facts:** - Wind players who received self-regulation instruction — including specific written practice goals reviewed at the next lesson — improved practice quality at moderate-to-large effect sizes ( $d \approx 0.4$ – $0.6$ ) over students given generic practice instructions. - Practice quality gains were measurable on observational rubrics, not just self-report. - The intervention was structured: students wrote down specific session targets, the procedures they would use, and a brief reflection after.

**DOK 2 — Summary:** Miksza ran the cleanest experiment on assignment specificity. Tell a student to "practice the Bach" and you get whatever they felt like doing that week. Hand them a written target with a specific procedure and practice quality jumps half a standard deviation. A real lever. Almost entirely missing from what most teachers actually send home.

**Link:** [Miksza publications](#) (paywalled at Sage; related open-access overview at [PMC: Setting the Stage for SRL Instruction in Musical Practice](#))

## **1.2 Challenging — Creech & Hallam on controlling parental involvement**

**Source:** Creech, A., & Hallam, S. (2003). Parent–teacher–pupil interactions in instrumental music tuition: A literature review. *British Journal of Music Education*, 20(1), 29–44.

**DOK 1 — Facts:** - Parental involvement in practice has *opposite* effects depending on style. Autonomy-supportive parents predicted persistence; controlling parents (correcting every error, demanding repetition) predicted dropout. - The amount of parental supervision was a weaker predictor than its style. - Even with clear teacher instructions, parents could turn structured assignments into controlling environments that undermined motivation.

**DOK 2 — Summary:** This is the inconvenient finding for the information-first thesis. Creech and Hallam showed that giving parents specific instructions doesn't always help. Sometimes it gives them ammunition to over-control. Better information delivered by an over-involved parent can produce less practice and faster dropout. The relay isn't neutral. It changes what the information becomes.

**Link:** [Creech & Hallam \(2003\), ResearchGate \(abstract\)](#) — paywalled at [Cambridge Core](#)

## **1.3 Emerging — Andreou on case-study evidence of assignment specificity**

**Source:** Andreou, E. (2022). *The relationship between private studio-based piano lessons and home-based private practice: Case studies of young piano students*. Doctoral thesis, UCL Institute of Education.

**DOK 1 — Facts:** - Students whose teachers gave specific written notes during the lesson and referenced what to practice at home produced more structured home-practice sessions than students whose teachers did not. - Without specific direction, even successful in-lesson performances did not translate to productive home practice. Students could not reconstruct the lesson on their own. - Lower-level students were the most dependent on teacher-provided structure to begin a practice session at all.

**DOK 2 — Summary:** Andreou's case studies are small and qualitative, but they're directly on the population this BrainLift is about. They show what Miksza's numbers look like in real living rooms. A great lesson didn't carry over on its own. Students who left with specific notes practiced differently. The ones who left with vague encouragement sat at the bench and froze.

Link: [Andreou \(2022\), UCL Discovery \(open-access thesis\)](#)

## Category 2: The Strategy Gap

*The second half of the BrainLift — that students don't know how to practice. Effective practice has identifiable behaviors, and novices default to none of them.*

### 2.1 Supporting — Duke et al. on the eight error-handling behaviors

**Source:** Duke, R. A., Simmons, A. L., & Cash, C. D. (2009). It's not how much; it's how: Characteristics of practice behavior and retention of performance skills. *Journal of Research in Music Education*, 56(4), 310–321.

**DOK 1 — Facts:** - 17 advanced piano students learned a 3-measure passage from Shostakovich's Piano Concerto No. 1. The top performers were not distinguished by total practice time. - Eight specific behaviors discriminated top from bottom: hands together early, inflection from the start, thoughtful pauses, errors preempted, errors addressed immediately, precise error identification, systematic tempo variation, and repeat-until-stable. - Three behaviors were used by *every* top-ranked pianist: precise error identification, deliberate tempo variation, and repeat-until-stable.

**DOK 2 — Summary:** Duke gave a difficult passage to 17 advanced pianists and watched what they did. The best ones weren't practicing more. They were practicing differently. They stopped on errors immediately, named what went wrong, fixed that one thing, then moved on. The worst ones played the passage start to finish over and over. The eight behaviors are the empirical answer to how effective practice works. Most beginner students do none of them.

Link: [Bulletproof Musician — accessible summary](#) (paywalled at [Sage](#))

### 2.2 Challenging — Macnamara on deliberate practice as insufficient

**Source:** Macnamara, B. N., Hambrick, D. Z., & Oswald, F. L. (2014). Deliberate practice and performance in music, games, sports, education, and professions: A meta-analysis. *Psychological Science*, 25(8), 1608–1618.

**DOK 1 — Facts:** - Across 88 studies, deliberate practice explained ~21% of variance in music performance — far less than the 10,000-hours framing implied. - Variance attributable to deliberate practice was even smaller in education (~4%) and professions (<1%). - Other factors — starting age, working memory, instructional quality, and individual differences — accounted for the bulk of expert performance.

**DOK 2 — Summary:** Macnamara is the inconvenient finding for any "strategy is the lever" thesis. Deliberate practice only explains 21% of music performance. Teaching kids the eight Duke behaviors won't carry them as far as the field assumes. The strategy gap is real and closing it matters. But it's

not the whole story. The other 79% is starting age, working memory, motivation, instructor quality, and things students can't control.

**Link:** [Macnamara, Hambrick & Oswald \(2014\), open-access PDF](#)

### **2.3 Emerging — Suzuki & Mitchell on play vs non-play strategies**

**Source:** Suzuki, A., & Mitchell, H. F. (2022). What makes practice perfect? How tertiary piano students self-regulate play and non-play strategies for performance success. *Psychology of Music*, 50(4), 1171–1189.

**DOK 1 — Facts:** - Successful tertiary piano students used a mix of *playing* strategies (slow practice, segmentation, error correction) and *non-playing* strategies (silent score study, mental rehearsal, planning, reflection). - Non-playing work was a measurable fraction of total practice time, not an afterthought. - Less successful students did proportionally less non-playing work and more unstructured run-throughs.

**DOK 2 — Summary:** Suzuki and Mitchell extend Duke's eight behaviors into what advanced pianists actually do. The headline finding is that the best practice isn't all at the keyboard. Score study, mental rehearsal, planning. The field has no language for teaching any of it. Most beginners are told to practice for 30 minutes, which means 30 minutes of playing. Suzuki and Mitchell say the practice that matters most is half keyboard time and half something else entirely.

**Link:** [Suzuki & Mitchell \(2022\), Psychology of Music](#) — paywalled at Sage

## **Category 3: The Translation Gap**

*Why even good "what" and "how" don't reach the bench. Covers lesson-to-home transfer and the consistency / distribution claim.*

### **3.1 Supporting — Prichard on explicit practice instruction transferring to home**

**Source:** Prichard, S. (2021). The impact of music practice instruction on middle school band students' independent practice behaviors. *Journal of Research in Music Education*, 69(1), 77–94.

**DOK 1 — Facts:** - Middle-school band students who received explicit practice-strategy instruction during lessons changed their independent practice behaviors at home — measurably, on observation. - Students who didn't receive explicit instruction did not improve their home practice behaviors over the same period. - The effect held for specific behaviors taught (segmentation, slow practice, error targeting), not just general "practice more."

**DOK 2 — Summary:** Prichard answers the most important question in this BrainLift. If you teach a kid how to practice, do they actually do it differently at home on their own? Yes, but only when the

instruction is explicit. "Practice 30 minutes" doesn't transfer. "Stop on errors and isolate them" does. The strategy makes the trip home when it's taught directly. Otherwise nothing does.

**Link:** [Prichard \(2021\), JRME](#) — paywalled at Sage

### **3.2 Challenging — Hanus & Fox on gamification backfiring**

**Source:** Hanus, M. D., & Fox, J. (2015). Assessing the effects of gamification in the classroom: A longitudinal study on intrinsic motivation, social comparison, satisfaction, effort, and academic performance. *Computers & Education*, 80, 152–161.

**DOK 1 — Facts:** - Students in a gamified course (badges, leaderboards, points) were compared to students in a non-gamified version of the same course over 16 weeks. Both groups started at equivalent motivation levels. - The gamified group showed *declining* intrinsic motivation, satisfaction, and empowerment over the semester. - Final exam scores were significantly *lower* in the gamified group, mediated by the decline in intrinsic motivation.

**DOK 2 — Summary:** This is the inconvenient finding for the dominant translation tool on the market. Every piano app uses gamification to bridge the lesson-to-home gap. Points, streaks, badges. Hanus and Fox ran it over 16 weeks. The gamified students finished with lower motivation and lower scores. Same content. Worse outcome. The tool the industry sells to solve the gap doesn't translate. It corrodes.

**Link:** [Hanus & Fox \(2015\), ScienceDirect](#) — paywalled at Elsevier

### **3.3 Emerging — Karpicke on retrieval and spacing**

**Source:** Karpicke, J. D., & Roediger, H. L. (2008). The critical importance of retrieval for learning. *Science*, 319(5865), 966–968. Plus: Karpicke, J. D., & Bauernschmidt, A. (2011). Spaced retrieval: Absolute spacing enhances learning regardless of relative spacing. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 37(5), 1250–1257.

**DOK 1 — Facts:** - Repeated retrieval over distributed time produced ~200% better long-term retention than repeated retrieval with no spacing between sessions. - The benefit came from absolute spacing — the total time between retrieval events — not from any particular schedule (expanding, equal, or otherwise). - The effect generalizes across declarative learning, motor learning, and skill acquisition.

**DOK 2 — Summary:** Karpicke isn't a music researcher. He's a cognitive scientist whose work on retrieval and spacing is too robust to ignore. Short distributed sessions beat long massed ones at the same total time. By 200% in his data. The piano field hasn't tested this cleanly in beginners but the cognitive science is settled. Five days of 30 minutes beats one session of 150 minutes. The reason isn't motivation. It's how the brain consolidates skill.

Link: [Karpicke & Roediger \(2008\), Science](#); [Karpicke Lab open-access PDF](#)

## Category 4: The Dropout Consequence

*What happens when home practice fails. The cost — visible in the dropout rate that ends most piano careers.*

### 4.1 Supporting — McPherson on commitment and the failure spiral

**Source:** McPherson, G. E. (2005). From child to musician: Skill development during the beginning stages of learning an instrument. *Psychology of Music*, 33(1), 5–35. Plus: Evans, P., McPherson, G. E., & Davidson, J. W. (2013). The role of psychological needs in ceasing music and music learning activities. *Psychology of Music*, 41(5), 600–619.

**DOK 1 — Facts:** - 157 beginning instrumentalists were tracked over three years. Pre-learning *commitment* — the child's stated intended duration of playing — predicted achievement more strongly than weekly practice minutes. - Children who quit had typically experienced months of unproductive practice, repeated lesson failure, and declining self-efficacy *before* the family decided to stop. - The "failure spiral" framing: bad home practice → lesson failure → eroded self-efficacy → protective disengagement → quitting.

**DOK 2 — Summary:** McPherson tracked 157 kids for three years. Quitting isn't sudden. It's the end of a chain that started months earlier with practice that produced nothing. The child stopped trying long before the parent canceled lessons. Unproductive practice doesn't just slow progress. It kills the belief that progress is possible.

Link: [McPherson \(2005\), Psychology of Music](#) — paywalled at Sage; [Evans, McPherson & Davidson \(2013\), open-access PDF on SDT site](#)

### 4.2 Challenging — Mosing et al. on the heritability of practice

**Source:** Mosing, M. A., Madison, G., Pedersen, N. L., Kuja-Halkola, R., & Ullén, F. (2014). Practice does not make perfect: No causal effect of music practice on music ability. *Psychological Science*, 25(9), 1795–1803.

**DOK 1 — Facts:** - Twin study with  $N > 10,000$  Swedish twins. Heritability of music practice itself was estimated at 40–70%. - Within monozygotic twin pairs (genetically identical), the twin who practiced more was *not* reliably more musically able than the twin who practiced less. - The correlation between practice and ability was largely explained by shared genetic factors influencing both.

**DOK 2 — Summary:** Mosing is the finding nobody in music education wants to talk about. The willingness to practice, the focus, the sustained interest, is itself partly genetic. So is the ability practice is supposed to build. When you control for the genetic confounding, the causal claim that

practice produces ability gets weaker. For dropout this matters. Kids who quit may not be quitting because their practice was bad. They may be quitting because the capacity to stick with practice was never there.

**Link:** [Mosing et al. \(2014\), Psychological Science](#) — [open-access PDF on author's page](#)

### **4.3 Emerging — Costa-Giomi on early predictors of dropout**

**Source:** Costa-Giomi, E. (2004). "I do not want to study piano!" Early predictors of student dropout behavior. *Bulletin of the Council for Research in Music Education*, 161/162, 57–62.

**DOK 1 — Facts:** - Future dropouts could be identified within the first weeks of beginning piano lessons by missed lessons, less practice time, and weaker progress. - Teacher-rated achievement was already lower for future dropouts before the decision to quit was made. - Demographic factors (income, parent education) predicted dropout less reliably than these early behavioral signals.

**DOK 2 — Summary:** Costa-Giomi found that teachers can see it coming within weeks. Missed lessons. Less practice. Weaker progress. The student has already disengaged before the family realizes. By the time the decision to quit happens, the kid left months ago. The signal is there, early, and almost nobody is using it.

**Link:** [Costa-Giomi \(2004\), JSTOR](#) — paywalled

## **DOK 3 — Insights**

### **Insight 1: The Problem Isn't What They Know, It's What Reaches the Bench**

Miksza (2015) proved specific assignments change practice quality. Prichard (2021) proved explicit strategy instruction transfers home. But Creech & Hallam (2003) showed the relay between lesson and home isn't neutral. The information exists. The strategies work. The breakdown isn't knowledge. It's delivery. A system that puts the right instruction at the bench at the moment of practice solves what better teaching alone can't, because better teaching still depends on an unreliable relay to get home.

### **Insight 2: The Industry's Fix Makes It Worse**

Duke (2009) identified the eight behaviors that separate effective practice from useless repetition. Hanus & Fox (2015) showed the tool the industry built to bridge the gap, gamification, actually lowers motivation and scores over 16 weeks. The field knows what good practice looks like. The market's

answer to getting students to do it makes them less likely to. The dominant translation mechanism is working against the science it claims to deliver.

### **Insight 3: The Spiral Starts Before Anyone Notices**

McPherson (2005) showed quitting is the end of a chain that starts with practice producing nothing. Costa-Giomi (2004) showed teachers can see the signal within weeks. Together: the dropout isn't a mystery. The warning signs are visible early and the cause is identifiable. Unproductive practice doesn't just fail to build skill. It actively destroys the belief that skill is buildable. The spiral is preventable if someone acts on the signal Costa-Giomi found, at the moment McPherson says the damage starts.

### **Insight 4: Strategy Works, But Not As Much As the Field Thinks**

Duke (2009) identified the behaviors. Prichard (2021) proved they transfer when taught explicitly. Macnamara (2014) showed deliberate practice only explains 21% of performance. Together: closing the strategy gap matters and it's teachable, but the field oversells it as the answer. The other 79% is things students can't control. A system built entirely on "teach them how to practice" will help and still leave most of the variance untouched. Honest product design starts from that ceiling, not from the assumption that strategy is the whole fix.

## **DOK 4 — Spiky Points of View (SPOVs)**

### **SPOV 1: The piano app industry built the translation layer. It's the wrong one.**

Every app from Simply Piano to Yousician tries to bridge the lesson-to-home gap with gamification. Duke (2009) identified the eight behaviors that make practice work. Hanus & Fox (2015) showed gamification lowers motivation and scores over 16 weeks. Miksza (2015) showed specific assignments move practice quality half a standard deviation. Prichard (2021) showed explicit strategy instruction actually transfers home. The science says what works: put the right instruction at the bench at the right moment. The industry built points, badges, and streaks instead. The dominant translation tool isn't just ineffective. It works against the science it claims to deliver.

The strongest counter is the engagement argument: without students opening the app, there's no instruction to deliver, and gamification is the only tool that reliably gets kids back. Hanus & Fox already ran that experiment. The gamified group had *higher* engagement and *lower* motivation and

scores than the non-gamified group on the same content over 16 weeks. Engagement bought attendance and lost the learning. The premise that engagement is the prerequisite confuses presence with progress.

## **SPOV 2: Every piano app is built to keep the student in the app. None is built to get them to the piano.**

Simply Piano, Yousician, Piano Marvel all optimize for session length, retention, and streaks because that's how they monetize. Duke (2009) showed what effective practice looks like. Hanus & Fox (2015) showed gamified engagement lowers motivation and scores. McPherson (2005) showed the student who isn't progressing is already on the path to quitting. Costa-Giomi (2004) showed teachers can see that signal within weeks. The app thinks it's winning because the student is still logged in. The student is already gone. Engagement in the app isn't progress on the piano. The business model and the learning outcome are pointing in opposite directions.

The strongest counter is the transfer argument: skills built in-app transfer to the real instrument, and time in the app *is* practice time. Acquilino & Scavone (2022) showed apps measure pitch and timing only — they can't see hands, posture, tone, or expression. The skills that transfer are the easy half. The skills that don't transfer — technique, musicality, physical control — are the half a teacher actually cares about. In-app fluency at note-matching is not piano fluency, and a student who is great at the app and frozen at the bench is the predictable result.

## **SPOV 3: Better teaching doesn't fix home practice. Better delivery does.**

Every solution in piano pedagogy assumes the problem is lesson quality. Train better teachers. Use better methods. Pick better repertoire. Miksza (2015) and Prichard (2021) showed the information and strategies exist and they work when they reach the student. Creech & Hallam (2003) showed the relay home isn't neutral. McPherson (2005) showed what happens when practice fails: the student quits. The lesson can be perfect. If the instructions don't survive the trip home in a form the student can use alone at the bench, the lesson didn't matter. The bottleneck isn't pedagogy. It's delivery.

The strongest counter is that pedagogy *is* delivery — a great teacher already does what this SPOV describes, with explicit strategy, specific assignments, and follow-up, and the real problem is teacher quality at scale rather than the channel itself. Even when the lesson is perfect, the relay home isn't neutral (Creech & Hallam 2003). The student practices alone. The teacher isn't there. Pedagogy ends when the student walks out the door. Delivery is what survives that exit. They are not the same problem and they don't have the same solution.

# Experts

## 1. Robert A. Duke — Supporting

- **Who:** Professor of Music and Human Learning, University of Texas at Austin; Director of the Center for Music Learning.
- **Focus:** Behavioral analysis of musicians' practice. The 2009 study "It's Not How Much; It's How" identified eight specific error-handling behaviors that distinguished the best advanced pianists from the rest, independent of total practice time.
- **Why Follow:** Duke's eight behaviors are the empirical anchor for the second belief — students don't know *how* to practice. He proves that effective practice is observable, identifiable, and not reducible to time on the bench.
- **Where:** [Duke, Simmons & Cash \(2009\), Journal of Research in Music Education](#)

## 2. Peter Miksza — Supporting

- **Who:** Professor of Music Education, Indiana University Jacobs School of Music.
- **Focus:** Self-regulation interventions in instrumental practice, with measurable effects on practice quality.
- **Why Follow:** Miksza's intervention work shows that specific written practice goals — reviewed at the next lesson — improve practice quality at  $d \approx 0.4$ – $0.6$  over generic instructions. That is the strongest empirical evidence for the first belief: that students don't know *what* to practice and that fixing the assignment closes part of the gap.
- **Where:** [Miksza \(2012\), JRME](#); [Miksza \(2015\), Psychology of Music](#)

## 3. Susan Hallam — Supporting

- **Who:** Emerita Professor of Education and Music Psychology, UCL Institute of Education.
- **Focus:** Practice strategies across expertise levels. Established that novices and experts practice qualitatively differently — novices run pieces start to finish without segmentation, error correction, or self-monitoring regardless of time invested.
- **Why Follow:** Hallam's strategy taxonomy is the foundational evidence that the strategy gap is real, persistent, and not solvable by trying harder. Her work is cited every time the field discusses what novice practice actually looks like.
- **Where:** [Hallam \(1997\), in Jørgensen & Lehmann \(Eds.\), Does Practice Make Perfect?](#)

## 4. Graham Fitch — Supporting (practitioner)

- **Who:** UK piano teacher, author, and founder of the Practising the Piano online academy and ebook series.
- **Focus:** Public, practitioner-facing methodology of practice — slow practice, segmentation, hands-separate work, mental practice, problem isolation, practice planning.
- **Why Follow:** Fitch is the closest the practitioner world comes to a working embodiment of this BrainLift's thesis. His entire career is teaching pianists *how* to practice, not what to play. He is the proof that the academic findings on strategy translate into a real, replicable pedagogy.
- **Where:** [Practising the Piano](#); [grahamfitch.com](http://grahamfitch.com)

## 5. Samantha Prichard — Emerging

- **Who:** Music education researcher.
- **Focus:** Whether explicit practice-strategy instruction transfers from the lesson into students' independent home practice.
- **Why Follow:** Prichard's 2021 study showed that explicit practice instruction changed middle-school instrumentalists' independent practice behaviors — direct empirical evidence that the lesson-to-home translation problem is solvable when teachers actually teach the strategy. She closes the loop the rest of the supporting evidence opens.
- **Where:** [Prichard \(2021\), JRME](#)

## 6. Edward L. Deci & Richard M. Ryan — Challenging

- **Who:** Founders of Self-Determination Theory (SDT); University of Rochester (and global affiliations).
- **Focus:** Motivation psychology. Their meta-analytic and theoretical work argues that intrinsic motivation is the engine of sustained engagement, that tangible rewards undermine it, and that controlling environments backfire — especially in children.
- **Why Follow:** Deci and Ryan are the steelman of the motivation camp. The third belief of this BrainLift — that the problem isn't motivation — has to survive their work to be defensible. Featuring them is the honest version of engaging the strongest counter-argument rather than dismissing a strawman.
- **Where:** [Deci, Koestner & Ryan \(1999\), Psychological Bulletin](#); [Ryan & Deci \(2017\), Self-Determination Theory](#)

## 7. Andrea Creech — Challenging

- **Who:** Professor and music education researcher, McGill University Schulich School of Music.
- **Focus:** The social and relational structure of music learning — parental involvement, peer learning, and the conditions under which students internalize practice habits.
- **Why Follow:** Creech offers the second steelman: practice fails not from missing information but from missing social structure. Her work on parental involvement (with Hallam) shows controlling supervision predicts dropout and that the *style* of social support matters more than its presence. Forces this BrainLift to defend why information-and-skill is the right lever rather than the social environment around the bench.
- **Where:** [Creech & Hallam \(2003\), British Journal of Music Education](#); [Boucher, Creech & Dubé \(2021\), Psychology of Music](#)

## 8. Jeffrey D. Karpicke — Supporting (cross-domain)

- **Who:** Professor of Psychological Sciences, Purdue University; head of the Cognition and Learning Laboratory.
- **Focus:** Retrieval practice and the spacing effect. Established that repeated retrieval over distributed time produces dramatically more long-term learning than massed restudy, with spaced retrieval producing up to a 200% improvement in retention vs. unspaced repetition.
- **Why Follow:** Karpicke is the cross-domain anchor for the consistency claim — that short, distributed sessions outperform long, massed ones at the same total time. The piano field hasn't tested distribution cleanly in beginners, but Karpicke's retrieval-and-spacing data is so robust across domains that any home-practice design that ignores it is fighting cognitive science. He gives this BrainLift the evidence to argue that consistency isn't a vibe — it's a measurable mechanism.
- **Where:** [Karpicke & Roediger \(2008\), Science](#); [Karpicke & Bauernschmidt \(2011\), JEP:LMC](#); [Karpicke & Blunt \(2011\), Science](#)

## Key Assumptions

- **Population:** This BrainLift addresses students who are already taking piano lessons (private, group, or studio) and struggle to practice consistently at home between lessons. It does not address self-taught beginners with no instructor. **Fail:** if the same practice-failure patterns show up identically in self-taught learners with no teacher in the loop, the lesson-to-home translation framing collapses and the thesis would need to be rewritten around something more general than home practice between lessons.

- **Lesson quality:** The lesson itself is assumed to be competent. The breakdown happens downstream — between the lesson and the bench — not inside the lesson. **Fail:** if practice failure is dominated by upstream pedagogy problems (poor teacher training, wrong repertoire selection, bad sequencing), then the delivery thesis is treating a symptom and the real lever is teacher development.
- **Generalization of the eight Duke behaviors:** The behaviors come from N=17 *advanced* pianists. This BrainLift assumes they generalize to beginners — that the same error-handling, segmentation, and tempo-variation behaviors that distinguish top advanced practicers also distinguish productive from unproductive beginner practice. **Fail:** if beginners' practice quality is dominated by behaviors not on Duke's list (basic note-reading, hand position, motor coordination), the strategy gap is differently shaped at the start.
- **Cross-domain transfer of spacing:** Karpicke's retrieval-and-spacing data comes from declarative learning. This BrainLift assumes it transfers to motor and musical skill acquisition. The cognitive science generally supports this, but no piano-specific RCT has tested distributed vs. massed schedules with total minutes held constant. **Fail:** if music skill behaves more like procedural-only learning that benefits from massed practice (some motor learning literature suggests this for short windows), the consistency claim weakens.
- **Macnamara ceiling applies here:** The 21% deliberate-practice ceiling is across all music populations including elite. This BrainLift assumes the ceiling is meaningful for the beginner / intermediate population it targets. **Fail:** if the beginner-specific ceiling is much higher (because basic skills are more practice-responsive than expert skills), Insight 4 overstates the limit.
- **Practice quality is observable at home:** SPOV 3 assumes the eight Duke behaviors and Miksza-style assignment-following can be inferred from audio, session logs, or lightweight signals at the bench. **Fail:** if the behaviors require visual observation of hands or in-person teacher presence to be reliably measured, the delivery-layer thesis needs hardware not yet specified.

## Self-Critique

- [x] Every SPOV has a named counter-argument and an honest rebuttal.
- [x] Every insight traces to DOK 2 sources with years.
- [x] Each Knowledge Tree category has Supporting, Challenging, and Emerging entries.
- [x] Each featured expert maps to a specific claim, not generic relevance.
- [x] In Scope bullets connect to Knowledge Tree categories: *what* (Cat 1), *how* (Cat 2), *translation + consistency* (Cat 3), *cost* (Cat 4). Motivation counter is integrated as Challenging entries throughout, per the guide.
- [x] Document reads standalone — no internal references, no project-specific shorthand.

## Honest gaps

- **Consistency / distribution evidence is thin in the music-specific literature.** Karpicke and Roediger come from cognitive science; the cross-domain transfer is the assumption (flagged above). No piano-specific RCT in beginners with total minutes held constant. The In Scope bullet stays in because the cognitive science is robust, but the BrainLift's strongest claim on consistency is "the field hasn't tested this cleanly," not "the field has tested this and we know the answer."
- **Motivation school is steelmanned but not exhaustively engaged.** Deci & Ryan and Creech are featured as challengers, but the motivation literature in piano specifically (Evans, Bonneville-Roussy, Davidson) only surfaces in passing rather than getting its own dedicated rebuttal. A motivation-camp reviewer could argue the BrainLift moved the goalposts: redefining "motivation" as something that flows from competence, then claiming the problem isn't motivation.
- **Western / Anglophone bias.** Almost every cited study comes from US, UK, Australian, or Northern European samples. Cultural variation in practice habits, parental involvement style, and the meaning of music education isn't addressed.
- **Adult learner population is under-represented.** Suzuki & Mitchell (2022) on tertiary students is the closest evidence for adult applicability. Most of the dropout, parental-involvement, and lesson-to-home translation literature is on children. Whether the same delivery thesis holds for adult returners is asserted, not proven.
- **No data on what the delivery layer should actually look like.** This BrainLift establishes that delivery is the bottleneck. It does not establish whether the right delivery is software, a teacher portal, an AI tutor, written lesson cards, video review, or something else. The product question stays open by design.
- **The 21% ceiling cuts both ways.** Insight 4 names the ceiling honestly. But the same finding partly undercuts the BrainLift's own urgency: if strategy only buys 21%, the spike against the gamification industry is moderated — they're solving a different 79% poorly, but the strategy camp can only solve their 21% well.